

O'REILLY®

Advanced Microsoft Power BI

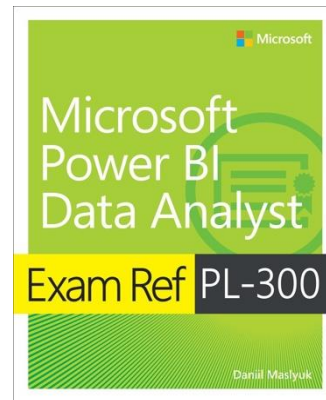
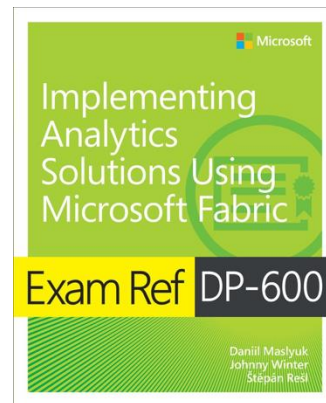
Presented by
Daniil Maslyuk



Daniil Maslyuk

Your presenter today

- Independent consultant
- Author of the official Power BI exam refs
- Blog: xxlbi.com
- LinkedIn: [daniilmaslyuk](https://www.linkedin.com/in/daniilmaslyuk)
- X: [@DMaslyuk](https://twitter.com/DMaslyuk)



Available through Safari!



Your Power BI experience

How long have you been using Power BI for?

- Less than 1 month
- $1 \text{ month} \leq X < 3 \text{ months}$
- $3 \text{ months} \leq X < 1 \text{ year}$
- $1 \text{ year} \leq X < 2 \text{ years}$
- $2 \text{ years} \leq X < 5 \text{ years}$
- 5 years or more



Course overview

- Lesson 1: External tools
- Lesson 2: Advanced data modeling concepts
- Lesson 3: Other advanced techniques



Lesson 1

External tools

- DAX Studio
- Tabular Editor
- ALM Toolkit
- VertiPaq Analyzer



Lesson 1

Exercise: use external tools

- DAX Studio
 - Open Adventure Works.pbix
 - Use Performance analyzer to identify the slowest visual
 - Copy the DAX query of the slowest visual into DAX Studio
 - Enable server timings and count storage engine (SE) queries
- Tabular Editor
 - Create new measures from columns
<https://docs.tabulareditor.com/te2/Useful-script-snippets.html>
- ALM Toolkit
 - Compare your current .pbix file and Reference.pbix



Lesson 2

Advanced data modeling concepts

- Dynamic format strings and calculation groups
- Row- and object-level security
- Composite models

Lesson 2

Exercise: use advanced data modeling techniques

- Create a calculation group to switch between
 - The default Sales and Date relationship and
 - The relationship based on Sales[ShipDateKey] and Date[DateKey]
- Create a row-level security (RLS) role based on Customer[Country-Region] and test it
- If you have a Power BI service account: create a composite model
 - Publish Adventure Works.pbix to the Power BI service
 - Connect to the Adventure Works semantic model
 - Add Targets.csv to your local data model
 - Create relationships:
 - Targets[End of Month] and Date[Date]
 - Targets[Category] and Product[Category]
 - Compare actual and target sales amounts



Lesson 3

Other advanced techniques

- Embedding images in themes
- Field parameters
- Enabling visual personalization

Lesson 3

Exercise: use advanced techniques to improve user experience

- Embed a background image in a theme
 - Encode Background.png
 - Example website for encoding: www.base64-image.de
 - Add the base64 code Theme - template.json
 - Add the new theme to your .pbix file
- Enable visual personalization with a perspective
 - Create a perspective with fields of your choice
 - Add at least one explicit measure to the perspective
 - Test your perspective in Excel or the Power BI service

Thank you!

